Project Design Phase-II
Technology Stack (Architecture & Stack)

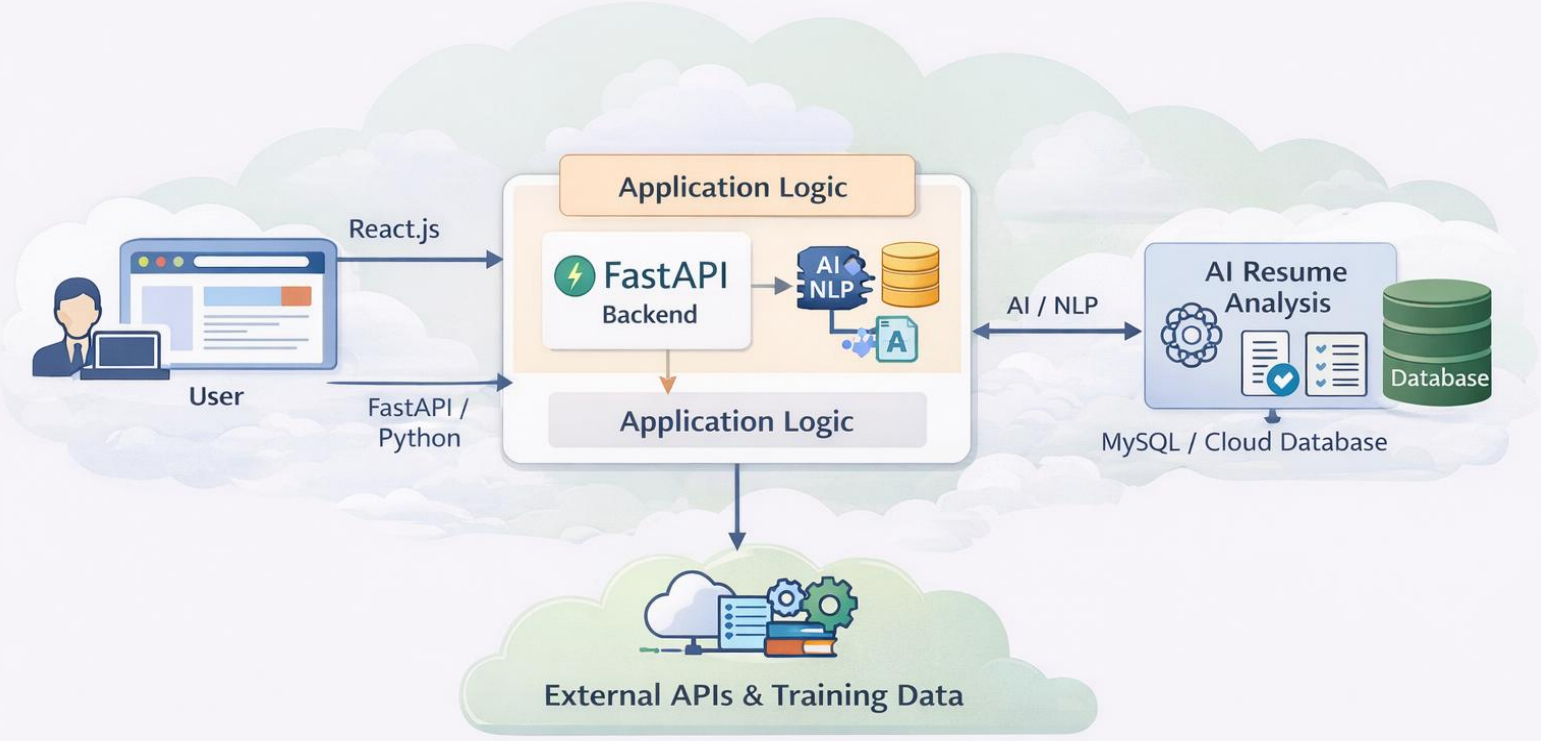
Date	31 January 3035
Team ID	699ec790745871b04830f7fe
Project Name	VidyaMitra: The Intelligent Career Agent AI-Powered Resume Evaluator, Trainer & Career Agent
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: VidyaMitra: The Intelligent Career Agent AI-Powered Resume Evaluator, Trainer & Career Agent

Technical Architecture Diagram



Technical Architecture Diagram – VidyaMitra System

Sr. No.	Components	Discription	Technology
1.	User Interface	Provides interactive interface for users to upload resumes, view analysis, take mock interviews, and track progress.	React.js / HTML / CSS / JavaScript
2.	Application Logic-1	Handles user registration, login authentication, resume upload, and dashboard operations.	Python (FastAPI Backend)
3.	Application Logic-2	Processes resume parsing and AI-based resume analysis to extract skills and experience.	Python AI / NLP Libraries
4.	Application Logic-3	Generates skill gap analysis, training recommendations, and mock interview questions.	AI Model / Recommendation Algorithm
5.	Database	Stores user information, resumes, analysis results, interview results, and training data.	MySQL / PostgreSQL
6.	Cloud Database	Cloud-based storage for scalability and remote access to data.	Firebase / MongoDB Atlas / Cloud Database
7.	File Storage	Stores uploaded resumes and related documents securely.	Local File System / Cloud Storage
8.	External API-1	Provides AI-based language processing and resume evaluation support.	OpenAI API / NLP APIs
9.	External API-2	Provides learning resources or training recommendations.	Course Recommendation APIs
10.	Machine Learning Model	AI model used to analyze resumes, extract skills, detect skill gaps, and provide personalized career recommendations.	NLP-Based Resume Analysis Model using Python (spaCy, Scikit-learn, Transformers)
11.	Infrastructure (Server / Cloud)	Platform where the application is deployed and executed for users. Local Server Configuration: Used during development and testing of the VidyaMitra system. Cloud Server Configuration : Cloud deployment to make the application accessible online and scalable.	Localhost Server, Python FastAPI, Node.js, MySQL AWS / Google Cloud / Heroku / Docker / Kubernetes

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Performance The VidyaMitra platform uses open-source frameworks to build the frontend interface, backend services, and AI-based resume analysis system. These frameworks allow flexibility, faster development, and community support.	React.js, FastAPI (Python), Node.js, NLP Libraries (spaCy, NLTK)
2.	Security Implementations	The system implements authentication and data protection mechanisms to ensure that user resumes and personal information remain secure. Access control, encrypted passwords, and secure API communication are used.	JWT Authentication, HTTPS Encryption, SHA-256 Password Hashing, OWASP Security Practices
3.	Scalable Architecture	The system follows a **3-tier architecture** consisting of presentation layer, application layer, and database layer. This architecture supports scalability and allows the platform to handle multiple users and resume analyses efficiently.	React Frontend, FastAPI Backend, MySQL / Cloud Database
4.	Availability	The application is designed to be available to users continuously with minimal downtime. Cloud deployment and distributed server infrastructure ensure the platform remains accessible for career guidance services.	Cloud Hosting (AWS / Google Cloud / Heroku), Load Balancing
5.	Performance	The system is optimized to process resume analysis quickly using efficient algorithms and backend processing. Caching and optimized database queries help improve response time and handle multiple requests effectively.	FastAPI Async Processing, Database Optimization, Redis Cache (Optional)